

Advanced approximate and symbolic computing

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Function interpolation and approximation I.

Interpolation

- We know the values (and maybe the values of the derivative) of a function $f(x)$ on the interval $[a, b]$ at finite many points (which are called *interpolation base-points*).
- We want to find a function $p(x)$ such that it equals to $f(x)$ in the base points:

$$p(x_i) = f(x_i) \quad (i = 1, \dots, n)$$

- Depending on the properties of $p(x)$ we have several methods for interpolation.

Lagrange interpolation

Let the base points x_i are given (pairwise different) and their corresponding function values $f_i \stackrel{\text{def}}{=} f(x_i)$, where $i = 1, \dots, n$.

The Lagrange polynomial $P_n(x)$ of degree $n - 1$ which interpolates the function $f(x)$ can be calculated using the formula

$$P_n(x) = \sum_{i=1}^n f_i L_i(x)$$

where

$$L_i(x) \stackrel{\text{def}}{=} \frac{(x - x_1) \dots (x - x_{i-1})(x - x_{i+1}) \dots (x - x_n)}{(x_i - x_1) \dots (x_i - x_{i-1})(x_i - x_{i+1}) \dots (x_i - x_n)}.$$

Theorem. This polynomial uniquely exists.

Trigonometric interpolation

Trigonometric interpolation

- In the followings, we assume that $f(x)$ is 2π -periodic function¹, and we want to find its interpolation on the interval $[0, 2\pi)$.
- According to the general theory (see lecture notes) the function set

$$\{1, \sin x, \cos x, \sin 2x, \cos 2x, \dots, \sin mx, \cos mx\}$$

is of Chebyshev-type on this interval, hence the interpolation polynomial uniquely exists.

- We fix an **even** number of base points and their corresponding function values.

¹ $f(x + 2\pi) = f(x), \quad x \in \mathcal{R}$

Trigonometric interpolation

Given $1 \leq n = 2m + 1$ natural number, pairwise different *interpolation base points* $\{x_0, x_1, x_2, \dots, x_{n-1}\} \in [0, 2\pi)$ and function values $\{f_0, f_1, f_2, \dots, f_{n-1}\}$.

Find

$$T_n(x) = \frac{a_0}{2} + \sum_{l=1}^m (a_l \cos lx + b_l \sin lx)$$

trigonometric interpolation polynomial which satisfies

$$T_n(x_i) = f_i \quad (i = 0, 1, 2, \dots, n-1) \tag{1}$$

- We have chosen the base points to be real numbers, but they can be complex $f_l \in \mathcal{C}$.
- We are dealing only with equidistant case:

$$x_k = \frac{2\pi k}{n}, \quad k = 0, 1, 2, \dots, n-1.$$

- In the following formulas the notation ι **always** stands for

$$\iota = \sqrt{-1}$$

which is the imaginary unit.

Auxiliary formulas

- *Euler* formula:

$$e^{it} = \cos t + i \sin t,$$

from which the *Moivre-formula* follows:

$$e^{int} = \cos nt + i \sin nt.$$

- Moreover, since we have $x_k = 2\pi k/n$, then from the equation

$$e^{inx_k} = e^{in\frac{2\pi k}{n}} = e^{i2\pi k} = 1$$

it follows that

$$e^{-ilx_k} = e^{inx_k} e^{-ilx_k} = e^{i(n-l)x_k}$$

Auxiliary formulas

- Using these formulas we can derive

$$\cos lx_k = \frac{e^{ilx_k} + e^{i(n-l)x_k}}{2}, \quad (2)$$

$$\sin lx_k = \frac{e^{ilx_k} - e^{i(n-l)x_k}}{2i}, \quad (k = 0, 1, 2, \dots, n-1) \quad (3)$$

Phase polynomial

- We show that finding

$$T_n(x) = \frac{a_0}{2} + \sum_{l=1}^m (a_l \cos lx + b_l \sin lx)$$

can be reformulated as finding the so-called *phase-polynomial*

$$p_n(x) = \beta_0 + \beta_1 e^{ix} + \beta_2 e^{2ix} + \cdots + \beta_{n-1} e^{(n-1)ix} \quad (4)$$

which satisfies

$$p_n(x_k) = f_k, \quad k = 0, 1, 2, \dots, n-1 \quad (5)$$

- Note that $p_n(x)$ is not a polynomial in its traditional meaning.

Phase polynomial

$$T_n(x) = \frac{a_0}{2} + \sum_{l=1}^m (a_l \cos lx + b_l \sin lx) = \sum_{l=-m}^m \beta_l e^{ilx}$$

- Now, if we take the coefficients β_j as

$$\beta_0 = \frac{a_0}{2}, \quad \beta_j = \frac{1}{2}(a_j - ib_j), \quad \beta_{n-j} = \frac{1}{2}(a_j + ib_j), \quad j = 0, 1, 2, \dots, m \quad (6)$$

and as

$$a_0 = 2\beta_0 \quad a_l = \beta_l + \beta_{n-l}, \quad b_l = i(\beta_l - \beta_{n-l}), \quad l = 0, 1, 2, \dots, m, \quad (7)$$

then using the formulas from the previous **two** slides we can show that

$$f_k = T_n(x_k) = p_n(x_k), \quad k = 0, 1, 2, \dots, n-1,$$

meaning that $T_n(x)$ and $p_n(x)$ equal in the base points.

Using the substitutions $\omega \stackrel{\text{def}}{=} e^{ix}$ and $\omega_k \stackrel{\text{def}}{=} e^{ix_k}$, according to

$$p_n(x_k) = f_k, \quad k = 0, 1, 2, \dots, n-1$$

finding the phase-polynomial $p_n(x)$ is equivalent to the finding of the following Lagrange polynomial (over complex numbers)-

Find that (at most degree $n-1$) polynomial

$$P_n(\omega) = \beta_0 + \beta_1\omega + \dots + \beta_{n-1}\omega^{n-1}$$

which at the basis points

$$\omega_k = e^{ix_k}$$

takes the given function values f_k for all $k = 0, 1, \dots, n-1$.

- Due to the existence and uniqueness of Lagrange interpolation polynomial we have exactly one $P_n(\omega)$ polynomial.
- We are looking for simple and efficient method for finding the coefficients of $P_n(\omega)$.

Finding the coefficients

- Our previous statement can be re-phrased with formulas as

$$p_n(\omega_0) = f_0, p_n(\omega_1) = f_1, \dots, p_n(\omega_{n-1}) = f_{n-1},$$

which is in details:

$$\begin{aligned}\beta_0 + \beta_1\omega_0 + \dots + \beta_{n-1}\omega_0^{n-1} &= f_0 \\ \beta_0 + \beta_1\omega_1 + \dots + \beta_{n-1}\omega_1^{n-1} &= f_1 \\ &\vdots \\ \beta_0 + \beta_1\omega_{n-1} + \dots + \beta_{n-1}\omega_{n-1}^{n-1} &= f_{n-1}\end{aligned}$$

Finding the coefficients

- Note that $x_k = (2\pi k)/n$. In matrix-vector form we have

$$\begin{pmatrix} 1 & \omega_0 & \dots & \omega_0^{n-1} \\ 1 & \omega_1 & \dots & \omega_1^{n-1} \\ \vdots & \vdots & & \vdots \\ 1 & \omega_{n-1} & \dots & \omega_{n-1}^{n-1} \end{pmatrix} \begin{pmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_{n-1} \end{pmatrix} = \begin{pmatrix} f_0 \\ f_1 \\ \vdots \\ f_{n-1} \end{pmatrix}$$

- This is in compact form:

$$\mathbf{\Omega}\boldsymbol{\beta} = \mathbf{f}, \tag{8}$$

Finding the coefficients

Theorem 9.1 *The matrix Ω in the system (8) has the property*

$$\Omega^H \Omega = nI,$$

where $\Omega^H = \overline{\Omega}^T$.

Proof. See: lecture notes. ■

Finding the coefficients

- In order to solve the system (8) we can multiply with the inverse of matrix $\mathbf{\Omega}$.
- According to the previous theorem we have

$$\mathbf{\Omega}^{-1} = \frac{1}{n} \mathbf{\Omega}^H.$$

- The solution of the system (8) is

$$\boldsymbol{\beta} = \frac{1}{n} \mathbf{\Omega}^H \mathbf{f},$$

which has computational complexity $\mathcal{O}(n^2)$.

Note

The map $\mathcal{F} : \mathcal{C}^n \rightarrow \mathcal{C}^n$

$$(f_0, f_1, \dots, f_{n-1}) \mapsto (\beta_0, \beta_1, \dots, \beta_{n-1})$$

is called *discrete Fourier transformation*.